Recurrence Relations

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What is a Recurrence Relation?

- A recurrence relation defines a function in terms of its smaller inputs.
- Widely used to analyze recursive algorithms.

Examples:

$$T(n) = T(n-1) + 1$$

$$T(n) = 2T(n/2) + n$$

Divide and Conquer

Divide and Conquer Strategy:

- **Divide** the problem into smaller subproblems.
- **Solve** each subproblem recursively.
- **Combine** results to get the final solution.

Example: Merge Sort

- Divides array into two halves.
- Recursively sorts each half.
- Merges in linear time (n).

$$T(n) = 2T(n/2) + n$$

Why Solve Recurrences?

- To determine time complexity of recursive algorithms.
- To convert recursive definitions into closed forms.
- To compare algorithm efficiency.

Goal:

- Exact solution or asymptotic bound (Big-O, Θ).

Types of Recurrences

- **Linear Recurrence**

$$T(n) = T(n-1) + c$$

- **Divide and Conquer**

$$T(n) = aT(n/b) + f(n)$$

- **Non-standard**

$$T(n) = T(n/2) + T(n/3) + n$$

Methods to Solve Recurrences

- **Iteration (Unfolding) Method**
- **Substitution Method** (Induction)
- **Recursion Tree Method**
- **Master Theorem**

Iteration Method (Unfolding)

- Expand the recurrence step-by-step.
- Continue until base case is reached.
- Identify the pattern formed.
- Convert into summation and solve.

Steps in Iteration Method

Given:

$$T(n) = aT(n/b) + f(n)$$

1. Expand the recurrence.
2. Substitute repeatedly.
3. Stop at the base case.
4. Identify the pattern.
5. Form a summation.
6. Solve the summation.

Example 1

Solve:

$$T(n) = T(n-1) + 1$$

Expansion:

$$\begin{aligned} T(n) &= T(n-1) + 1 \\ &= (T(n-2) + 1) + 1 = T(n-2) + 2 \\ &= T(n-3) + 3 \\ &\cdots \\ &= T(1) + (n-1) \end{aligned}$$

$$T(n) = \Theta(n)$$

Example 2

Solve:

$$T(n) = 2T(n/2) + n$$

Expansion:

$$\begin{aligned} T(n) &= 2T(n/2) + n \\ &= 2(2T(n/4) + n/2) + n = 4T(n/4) + 2n \\ &= 4(2T(n/8) + n/4) + 2n = 8T(n/8) + 3n \\ &\dots \end{aligned}$$

After k steps:

$$T(n) = 2^k T(n/2^k) + kn$$

Example 2 (continued)

Stop when subproblem size is 1:

$$\frac{n}{2^k} = 1 \implies k = \log_2 n$$

Substitute k:

$$T(n) = 2^{\log_2 n} T(1) + n \log_2 n$$

$$T(n) = n \cdot T(1) + n \log_2 n$$

$$T(n) = \Theta(n \log n)$$

Example 3

Solve:

$$T(n) = T(n/2) + 1$$

Expansion:

$$\begin{aligned} T(n) &= T(n/2) + 1 \\ &= T(n/4) + 2 \\ &= T(n/8) + 3 \\ &\cdots \end{aligned}$$

For $n/2^k = 1$, we have $k = \log_2 n$:

$$T(n) = T(1) + \log_2 n$$

$$T(n) = \Theta(\log n)$$

Key Observations

- Recursion depth is often $\log n$ for divide and conquer.
- Total work depends on work per level.
- Common patterns:
  - Constant work per level gives $\log n$.
  - Linear work per level gives $n \log n$.

The Master Theorem: General Form

The Master Theorem determines the running time of divide and conquer algorithms directly from their recurrence relation without fully solving it.

General Form:

$$T(n) = aT\left(\frac{n}{b}\right) + \Theta(n^k \log^p n)$$

Constraints:

- $a \geq 1$
- $b > 1$
- $k \geq 0$
- p is a real number

Master Theorem: Cases

Given $T(n) = aT\left(\frac{n}{b}\right) + \Theta(n^k \log^p n)$:

Case 1: If $a > b^k$, then $T(n) = \Theta(n^{\log_b a})$

Case 2: If $a = b^k$

- **2.a:** If $p > -1$, then $T(n) = \Theta(n^{\log_b a} \log^{p+1} n)$
- **2.b:** If $p = -1$, then $T(n) = \Theta(n^{\log_b a} \log \log n)$
- **2.c:** If $p < -1$, then $T(n) = \Theta(n^{\log_b a})$

Case 3: If $a < b^k$

- **3.a:** If $p \geq 0$, then $T(n) = \Theta(n^k \log^p n)$
- **3.b:** If $p < 0$, then $T(n) = O(n^k)$

Problems (1 to 4)

Problem 1: $T(n) = 3T(n/2) + n^2$

Problem 2: $T(n) = 4T(n/2) + n^2$

Problem 3: $T(n) = T(n/2) + n^2$

Problem 4: $T(n) = 2^n T(n/2) + n^n$

Problems & Solutions (1 to 4)

Problem 1: $T(n) = 3T(n/2) + n^2$
$a = 3,\ b = 2,\ k = 2,\ p = 0 \Rightarrow \Theta(n^2)$ (Case 3.a)

Problem 2: $T(n) = 4T(n/2) + n^2$
$a = 4,\ b = 2,\ k = 2,\ p = 0 \Rightarrow \Theta(n^2 \log n)$ (Case 2.a)

Problem 3: $T(n) = T(n/2) + n^2$
$a = 1,\ b = 2,\ k = 2,\ p = 0 \Rightarrow \Theta(n^2)$ (Case 3.a)

Problem 4: $T(n) = 2^n T(n/2) + n^n$
$a = 2^n$ (not constant) $\Rightarrow$ Not applicable

Problems (5 to 8)

Problem 5: $T(n) = 16T(n/4) + n$

Problem 6: $T(n) = 2T(n/2) + n \log n$

Problem 7: $T(n) = 2T(n/2) + \frac{n}{\log n}$

Problem 8: $T(n) = 2T(n/4) + n^{0.51}$

Problems & Solutions (5 to 8)

Problem 5: $T(n) = 16T(n/4) + n$
$a = 16,\ b = 4,\ k = 1,\ p = 0 \Rightarrow \Theta(n^2)$ (Case 1)

Problem 6: $T(n) = 2T(n/2) + n \log n$
$a = 2,\ b = 2,\ k = 1,\ p = 1 \Rightarrow \Theta(n \log^2 n)$ (Case 2.a)

Problem 7: $T(n) = 2T(n/2) + \frac{n}{\log n}$
$a = 2,\ b = 2,\ k = 1,\ p = -1 \Rightarrow \Theta(n \log \log n)$ (Case 2.b)

Problem 8: $T(n) = 2T(n/4) + n^{0.51}$
$a = 2,\ b = 4,\ k = 0.51,\ p = 0 \Rightarrow \Theta(n^{0.51})$ (Case 3.b)

Problems (9 to 12)

Problem 9: $T(n) = 0.5T(n/2) + \frac{1}{n}$

Problem 10: $T(n) = 6T(n/3) + n^2 \log n$

Problem 11: $T(n) = 64T(n/8) - n^2 \log n$

Problem 12: $T(n) = 7T(n/3) + n^2$

Problems & Solutions (9 to 12)

Problem 9: $T(n) = 0.5T(n/2) + \frac{1}{n}$
$a = 0.5 < 1 \Rightarrow$ Not applicable

Problem 10: $T(n) = 6T(n/3) + n^2 \log n$
$a = 6,\ b = 3,\ k = 2,\ p = 1 \Rightarrow \Theta(n^2 \log n)$ (Case 3.a)

Problem 11: $T(n) = 64T(n/8) - n^2 \log n$
Negative term $\Rightarrow$ Not applicable

Problem 12: $T(n) = 7T(n/3) + n^2$
$a = 7,\ b = 3,\ k = 2,\ p = 0 \Rightarrow \Theta(n^2)$ (Case 3.a)

Problems (13 to 16)

Problem 13: $T(n) = 4T(n/2) + \log n$

Problem 14: $T(n) = 16T(n/4) + n!$

Problem 15: $T(n) = \sqrt{2}T(n/2) + \log n$

Problem 16: $T(n) = 3T(n/2) + n$

Problems & Solutions (13 to 16)

Problem 13: $T(n) = 4T(n/2) + \log n$
$a = 4,\ b = 2,\ k = 0,\ p = 1 \Rightarrow \Theta(n^2)$ (Case 1)

Problem 14: $T(n) = 16T(n/4) + n!$
$n!$ not standard $\Rightarrow \Theta(n!)$

Problem 15: $T(n) = \sqrt{2}T(n/2) + \log n$
$a = \sqrt{2},\ b = 2,\ k = 0,\ p = 1 \Rightarrow \Theta(\sqrt{n})$ (Case 1)

Problem 16: $T(n) = 3T(n/2) + n$
$a = 3,\ b = 2,\ k = 1,\ p = 0 \Rightarrow \Theta(n^{\log_2 3})$ (Case 1)

Problems (17 to 20)

Problem 17: $T(n) = 3T(n/3) + \sqrt{n}$

Problem 18: $T(n) = 4T(n/2) + cn$

Problem 19: $T(n) = 3T(n/4) + n \log n$

Problem 20: $T(n) = 3T(n/3) + \frac{n}{2}$

Problems & Solutions (17 to 20)

Problem 17: $T(n) = 3T(n/3) + \sqrt{n}$
$a = 3,\ b = 3,\ k = \frac{1}{2},\ p = 0 \Rightarrow \Theta(n)$ (Case 1)

Problem 18: $T(n) = 4T(n/2) + cn$
$a = 4,\ b = 2,\ k = 1,\ p = 0 \Rightarrow \Theta(n^2)$ (Case 1)

Problem 19: $T(n) = 3T(n/4) + n \log n$
$a = 3,\ b = 4,\ k = 1,\ p = 1 \Rightarrow \Theta(n \log n)$ (Case 3.a)

Problem 20: $T(n) = 3T(n/3) + \frac{n}{2}$
$a = 3,\ b = 3,\ k = 1,\ p = 0 \Rightarrow \Theta(n \log n)$ (Case 2.a)